The plant reproduction strategy studio

Teacher guide and worked answers

Distinguish sexual and asexual plant reproduction using process evidence rather than the number of visible parent plants.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsg/inputs/GROW SOW 2026-27.xlsx | GROW Weekly - Summer | C31

Describe sexual and asexual reproduction in plants.

New outcome; sexual reproduction is gamete fusion, not necessarily two separate parent plants. Original secondary-age exemplar at an upper-KS2 conceptual level; no qualification approval or completion claim.

Prior learning

Describe pollination and fertilisation.

Know that offspring inherit genetic information.

Success evidence

Identify whether gamete fusion occurs in a case.

Describe a runner or cutting as an asexual route.

Explain why sexual reproduction can involve one self-pollinating plant.

Equipment

Plant case cards, selected route, pencils; optional paper runner model.

Preparation

Use the case's stated reproductive process, not just appearance or number of plants.

No live cutting or rooting chemical is needed.

Practical care

Use paper models; no plant tasting, cutting tools or hormone powder.

Ready to teach alternative

All decisions use supplied cases and diagrams. A live propagation experiment is not required or claimed completed.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	One visible parent: one answer?
2	Retrieval	2-5	Recover the fusion event
3	I do	5-13	I look for gamete fusion

Slide	Stage	Minutes	Focus
4	I do	Within stage	I trace the runner route
5	I do	Within stage	I separate genetic similarity from appearance
6	We do	13-21	Nursery case conference
7	We do	Within stage	Choose a plan for the nursery
8	We do	Within stage	Challenge the shortcut rules
9	Hinge	21-24	A flower self-fertilises
10	You do	24-34	Your independent investigation
11	You do	Within stage	Use the evidence
12	You do	Within stage	Check before sharing
13	Review	34-38	Compare and improve
14	Review	Within stage	A question worth investigating
15	Exit	38-40	Show the science you can explain

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening One visible parent: one answer?

No: need process evidence. The lesson's central question is whether gametes fuse, not merely counting visible plants.

2 Retrieval Recover the fusion event

Gamete fusion; fertilised ovule; no. Do not treat every natural seed as necessarily sexual.

3 I do I look for gamete fusion

Think aloud with case B: I see only one parent plant, but the record says gametes fuse. Therefore it is sexual reproduction. A self-fertile plant can supply both gametes.

4 I do I trace the runner route

Shared I do time. Trace runner from parent to rooted node. The runner is a stem, not a root stretching between unrelated plants. It can remain attached during early growth.

5 I do I separate genetic similarity from appearance

Shared I do time. Use illustrative growth model, not measured trial data. Explain that mutation is a possible genetic change; supported pupils need only avoid the never-varies claim.

6 We do Nursery case conference

A/B sexual; C/D asexual. Pupils move labels or vote privately, then justify using the case wording.

7 We do Choose a plan for the nursery

Cuttings preserve the source’s inherited features more directly; seed crosses can introduce genetic variation. Keep suitable light/water/temperature consistent; clones do not guarantee identical growth or immunity.

8 We do Challenge the shortcut rules

Case B contradicts the first; environment/mutation limits second. Accept concise bounded explanations.

Reveal: Use the fusion evidence. A self-fertile plant can reproduce sexually; clone appearance can vary.

9 Hinge A flower self-fertilises

Sexual reproduction occurs because gametes fuse; it is not defined by the number of separate plants and is more than ordinary growth.

A. Sexual reproduction. - Gamete fusion is the defining event, even with one parent plant.

B. Asexual because only one parent plant is visible. - Counting plants misses the fusion event.

C. No reproduction, only taller growth. - A new offspring develops after fertilisation.

10 You do Your independent investigation

10 minutes across the three You do slides. Supply shared page 1 and one selected route page. All routes target the stated science objective; a labelled drawing or independent dictated explanation is valid.

11 You do Use the evidence

Shared You do time. Ask pupils which evidence supports their explanation. Read prompts or scribe without supplying the scientific conclusion.

12 You do Check before sharing

Shared You do time. Use the route-specific worked answers to identify one precise next step.

13 Review Compare and improve

4 minutes across Review. Offer partner or private teacher feedback. Ask for one specific revision linked to evidence; no public ranking.

14 Review A question worth investigating

Lundy cycle: invite written, spoken or pointed suggestions. Select one feasible question and explain how the pupil contribution changes the next example or investigation.

15 Exit Show the science you can explain

Fusion; runner/cutting; yes, gametes fuse. E4 asks why clones may look different.

Answers and assessment

R1-R3

Gamete fusion; fertilised ovule; growth develops one individual while reproduction produces offspring.

W1-W3

A/B sexual because gametes fuse; C/D asexual because no fusion. W2 cutting for retaining source genetic features, with controlled suitable growth conditions. W3 self-fertilisation is a one-parent sexual case; clones can differ through environment and sometimes mutation.

Hinge A-C

A correct; B incorrectly counts visible parents; C ignores production of offspring.

S1-S5

S1 sexual. S2 asexual. S3 gametes; runner. S4 cutting, because it develops vegetatively from the source and usually retains its genetic information. S5 different environment or mutation; an accurate example such as different light is sufficient.

N1-N4

A sexual/fusion; B sexual/fusion; C asexual/no fusion via runner; D asexual/no fusion via cutting. N2 gamete fusion defines the route. N3 cutting plus retained inherited features and relevant control. N4 genetically similar plants can respond differently to growth conditions; mutation can also alter inherited information.

T1-T4

T1 sexual uses fusion and typically produces genetic variation; asexual has no fusion and usually produces clones, e.g. runner/cutting. T2 self-fertilisation can use one plant yet remain sexual. T3 asexual seed formation according to the stated lack of fusion; this prevents treating all seeds as sexual. T4 cuttings usually retain source genes but appearance, establishment and disease outcomes depend on more than that; use suitable conditions and monitor health without guarantees.

E1-E4 / assessment

Fusion; accurate runner/cutting description; yes through gamete fusion; conditions or mutation may differ. Secure: correct process definition applied to self-fertilisation and vegetative cases, plus a bounded clone claim. Re-teach classification by highlighting the fusion evidence. Reflection accepts a relevant uncertainty.

Teaching and assessment notes

40 minutes: opening 2; retrieval 3; I do 8; We do 8; hinge 3; You do 10; review 4; exit 2. Zero-minute slides share their stage time.

Supply shared page 1, one selected route page 2, 3 or 4, and page 5. Do not require all three routes. An independent spoken explanation can be recorded verbatim by an adult.

The defining distinction is gamete fusion. Self-fertilisation remains sexual even if one plant supplies both gametes. Do not teach all seeds as universally sexual; optional asexual seed formation is explicitly signposted.

Paper runner and cutting cards support the same interactive decisions. A propagation practical would require separate ongoing care; this lesson does not claim roots appear within 40 minutes.

Access and responsive teaching

Supported

Sort stated gamete-fusion cases and explain one runner.

Standard

Compare processes and choose a nursery plan using evidence.

Stretch

Evaluate self-fertilisation, clone and all-seeds claims.

Regulation

Offer private writing, pointing, drawing or adult scribing. Allow pass-and-return for public questions and desk-based participation. Choose support from current learning evidence, not fixed ability labels.

Misconceptions to check

Sexual reproduction always needs two separate parent plants.

Some flowers can self-pollinate and self-fertilise; gamete fusion is still sexual reproduction.

Check: Ask whether gametes fuse in the case.

Every seed must be sexual and every clone looks identical.

Some plants form asexual seeds; environment can alter the appearance of genetically similar plants.

Check: Ask what process is actually stated.

Asexual offspring can never vary at all.

They are usually genetically very similar clones, but mutations and environmental effects can produce differences.

Check: Ask whether different growth conditions could change height.

Word	Meaning
sexual reproduction	Reproduction involving fusion of gametes.
asexual reproduction	Reproduction without gamete fusion.
runner	A horizontal stem that can produce a new plant at a node.
cutting	A piece of a parent plant used to grow a new plant.
clone	An offspring with the same inherited genetic information as its source, apart from changes such as mutations.
variation	Differences among individuals; genes and environment can contribute.

Scientific references

OpenStax Biology 2e: Pollination and fertilization

Checked September 2026. Pollination differs from gamete fusion; fertilised ovule develops into seed and ovary into fruit. College detail is selectively simplified for GROW.

OpenStax Biology 2e: Asexual reproduction

Checked September 2026. Runners, cuttings and clonal reproduction. Some plants can produce asexual seeds; core examples identify the actual process.

RHS: How plants reproduce

Checked September 2026. Sexual and clonal propagation overview; plants may use more than one reproductive strategy.